Beyond Passive Stifling: A Replication and Extension of the SIR-IM Model with Active Truth-Spreaders

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ABSTRACT

This project reproduces and extends the SIR-IM rumor spreading model[1], which incorporates an influence mechanism where the spreading probability depends on the number of spreaders. We re-implement the original mean-field equations and successfully replicate key results, including accelerated rumor diffusion and sensitivity to parameters such as individual opinion and forgetting rate. Building on this framework, we introduce a Truth-Spreader (T) state to model counter-rumor dynamics, capturing the interaction between misinformation and correction. Our extended model shows that active truth dissemination and correction mechanisms can significantly suppress rumor prevalence and reduce its duration. These findings confirm the validity of the original model while highlighting that its conclusions are limited by the absence of competing information processes. Overall, our work demonstrates the importance of incorporating counter-information dynamics for a more realistic understanding of information spreading.

Keywords: Social networks, SIR rumor spreading model, Influence mechanism

1 INTRODUCTION

Rumors are defined as unverified information or interpretations of public events that circulate through informal social channels. In the modern digital era, the rapid proliferation of online social platforms—such as WeChat, Weibo, and Twitter—has fundamentally altered the landscape of information diffusion. While these platforms facilitate efficient communication, they also serve as catalysts for the explosive spread of misinformation. A widespread rumor can lead to social instability, economic loss, and public panic. For instance, misinformation regarding public health crises or financial markets can trigger irrational collective behavior, making the study of rumor propagation dynamics a critical task for maintaining social order and purifying cyberspace.

To understand the mechanisms behind these phenomena, researchers have long drawn parallels between the spread of rumors and the transmission of biological viruses. Foundational research established that rumor spreading can be modeled using epidemic frameworks, such as the Susceptible-Infectious-Recovered (SIR) model. In these models, individuals transition from being credulous to spreaders upon contact, eventually becoming stiflers who no longer propagate the information. However, unlike simple diseases, rumor diffusion is deeply influenced by social psychology and collective reinforcement. Recent studies have moved beyond constant transmission rates to incorporate more complex behavioral mechanisms, such as time-dependent forgetting and the influence mechanism, where the probability of an individual believing a rumor increases with the number of active spreaders in their vicinity.

Among these modern frameworks, the SIR-IM model stands out for its incorporation of a dynamic spreading probability and individual resistance. While the SIR-IM model successfully captures the self-reinforcing nature of rumors, it relies on several simplifying assumptions. Most notably, it treats the stifling process as a passive event—either through forgetting or social saturation—and ignores the active role of counter-information. In real-world social networks, rumors do not spread in a vacuum; they are often met with active debunking efforts from official sources or fact-checkers. This absence of competing information dynamics represents a significant gap in the baseline model's ability to simulate realistic crisis management scenarios.

In this project, we provide a comprehensive reproduction and extension of the SIR-IM rumor spreading model. First, we replicate the core findings of the original study, while addressing a specific discrepancy regarding the unrealistic "spike" in early-stage growth observed in previous network-based simulations. Second, we introduce the SIR-CIM (Competitive Information Model), which incorporates a Truth-Spreader (T) compartment. This extension allows us to investigate the non-linear interaction between misinformation and active correction. Our results demonstrate that the timing of truth introduction is the primary determinant of suppression success, providing new insights into the strategic window for effective intervention in social networks.

2 MODEL FORMULATION AND ASSUMPTIONS

Research Question

The paper studies how rumors spread in online social networks, with a particular focus on how the number of active spreaders influences the overall spreading process. While rumor diffusion has often been modeled using epidemic frameworks, most classical models assume that the transmission rate is constant over time. This paper questions that assumption and asks whether the spreading rate should instead depend on how many people are currently spreading the rumor.

This question is important because real-world information diffusion does not behave like simple disease transmission. In social networks, the visibility and credibility of a rumor often increase as more people share it. For example, when many users repost the same message, others may be more likely to believe and further propagate it. Therefore, assuming a fixed transmission rate may fail to capture this kind of collective reinforcement effect, which plays a key role in online environments.

To address this issue, the paper proposes an extended SIR-type model that incorporates an "influence mechanism," where the probability of becoming a spreader depends on the current number of spreaders. In addition, the model includes time-dependent forgetting and individual-level resistance to rumors. The central research question can therefore be summarized as: how does incorporating these more realistic behavioral mechanisms change the dynamics of rumor spreading compared to traditional models?

Mathematical Model

Model Description

We adopt the SIR-IM rumor spreading model proposed by Qiu et al., which extends the classical SIR framework by incorporating an influence mechanism into the spreading dynamics.

The population is divided into three compartments:

- $C(t)$: Credulous individuals, who are susceptible to believing the rumor;
- $S(t)$: Spreaders, who actively propagate the rumor;
- $R(t)$: Stiflers, who no longer spread the rumor.

The system satisfies the normalization condition:

$$C(t) + S(t) + R(t) = 1. \tag{1}$$

Transition Mechanisms

The SIR-IM model introduces three key mechanisms that distinguish it from the classical SIR model.

1. **Influence-Based Spreading Probability**

   Unlike traditional models with a constant transmission rate, the SIR-IM model assumes that the spreading probability depends on the current number of spreaders.

   Let p denote the probability of contact between individuals, and let $m = S(t) \cdot N$ be the number of spreaders. The spreading and skeptical probabilities are defined as:

$$\alpha_1' = 1 - (1-p)^m, \quad \alpha_2' = (1-p)^m. \tag{2}$$

   These are normalized to obtain:

$$\alpha_1 = \frac{\alpha_1'}{\alpha_1' + \alpha_2' + p}, \quad \alpha_2 = \frac{\alpha_2'}{\alpha_1' + \alpha_2' + p}. \tag{3}$$

Here, α_1 represents the probability that a credulous individual becomes a spreader, while α_2 represents the probability that a credulous individual becomes a stifler. This formulation captures the influence mechanism, where the spreading probability increases with the number of spreaders.

2. **Time-Dependent Forgetting Mechanism**

   The model assumes that spreaders gradually lose interest in spreading the rumor over time. This is modeled through a time-dependent forgetting rate:

$$\beta_1(t) = 1 - e^{-\lambda t}, \tag{4}$$

   where $\lambda > 0$ controls the speed of forgetting. As time increases, $\beta_1(t)$ approaches 1, indicating that spreaders are increasingly likely to stop spreading the rumor.

3. **Interaction-Based Stifling Mechanism**

   When a spreader interacts with another spreader or a stifler, they may stop spreading the rumor with probability β_2. This represents social saturation or redundancy of information.

   Additionally, a fraction θ of credulous individuals may ignore the rumor entirely, representing an immune or indifferent behavior.

Mean-Field Dynamical System

Under the homogeneous mixing assumption, the dynamics of the system are described by the following nonlinear ordinary differential equations:

$$\frac{dC(t)}{dt} = -(\alpha_1 + \alpha_2 + \theta)C(t)S(t), \tag{5}$$

$$\frac{dS(t)}{dt} = \alpha_1 C(t)S(t) - \beta_1(t)S(t) - \beta_2(S(t) + R(t))S(t), \tag{6}$$

$$\frac{dR(t)}{dt} = (\alpha_2 + \theta)C(t)S(t) + \beta_1(t)S(t) + \beta_2(S(t) + R(t))S(t). \tag{7}$$

These equations describe the following processes:

- Rumor adoption: $C \to S$ at rate $\alpha_1 CS$;
- Skepticism or immunity: $C \to R$ at rate $(\alpha_2 + \theta)CS$;
- Forgetting: $S \to R$ at rate $\beta_1(t)S$;
- Social stifling: $S \to R$ through interactions at rate $\beta_2(S + R)S$.

Initial Conditions

The model assumes that initially only a small fraction of the population is spreading the rumor:

$$C(0) = \frac{N-1}{N}, \quad S(0) = \frac{1}{N}, \quad R(0) = 0. \tag{8}$$

Model Characteristics

Compared with the classical SIR model, the SIR-IM model has several important features:

- The spreading rate is dynamic and depends on the number of spreaders;
- The forgetting process is time-dependent;
- Individual heterogeneity is captured through skepticism and immunity;
- Social interaction effects are incorporated through the stifling term.

Key Assumptions

The model relies on several simplifying assumptions that make the system analytically tractable:

1. **Homogeneous mixing.** All individuals are assumed to interact uniformly at random, ignoring network structures such as clustering and heterogeneous connectivity.

2. **Collective influence.** The probability of adopting the rumor is assumed to increase with the number of spreaders, reflecting a reinforcement effect. This assumes that all spreaders contribute equally and independently.

3. **Time-dependent forgetting.** Individuals gradually lose interest in spreading the rumor over time, following a deterministic decay pattern. This ignores individual differences and external influences.

4. **Static individual resistance.** Each individual has a fixed probability of ignoring the rumor, independent of past exposure or social context.

Limitations of the Baseline Model

While the SIR-IM model provides a useful framework for capturing influence-based rumor spreading, it relies on several simplifying assumptions that limit its ability to describe more complex information dynamics in real-world social systems.

1. **Absence of counter-information dynamics.** The model focuses exclusively on the propagation of rumors and does not account for the presence of competing information. In real social networks, misinformation is often challenged by fact-checking, corrections, or authoritative sources. The lack of such counteracting mechanisms prevents the model from capturing the interaction between rumor and truth.

2. **Passive role of stiflers.** In the SIR-IM model, stiflers (R) are assumed to be inactive individuals who neither spread nor influence others. However, in practice, individuals who stop believing a rumor may actively correct others or spread factual information. Treating them as passive limits the model's ability to represent active intervention processes.

3. **Homogeneous influence assumption.** The model assumes that all spreaders exert identical influence on others, and that interactions occur uniformly across the population. This neglects heterogeneity in influence, such as highly connected or authoritative individuals who may play a disproportionate role in information diffusion.

4. **Deterministic behavioral dynamics.** Processes such as forgetting and resistance are modeled deterministically and uniformly across individuals. This ignores stochastic effects and individual-level variability, which can be significant in real-world rumor spreading.

Motivation for Model Extension

To address these limitations, we extend the baseline SIR-IM model by introducing an additional compartment representing *truth-spreaders* (T). This extension allows the model to explicitly capture the interaction between rumor and counter-information.

In the extended framework, truth-spreaders actively disseminate corrective information and may influence both credulous individuals and rumor spreaders. This enables the model to represent:

- the competition between rumor and truth,
- the correction of spreaders through exposure to factual information,
- and the dynamic balance between misinformation and its suppression.

By incorporating this additional mechanism, the model moves beyond a single-information diffusion process and becomes a coupled system of competing information dynamics, providing a more realistic description of information propagation in social networks.

3 OUR REPLICATION

In our study, we reproduce from the original SIR-IM model and extend it by introducing a competitive mechanism: the **Truth-Spreader**. We denote this extended framework as the **SIR-CIM model**. This addition allows us to observe how active debunking and official corrections counteract the diffusion of a rumor.

Replication Strategy

The primary objective of this study is to reproduce the key dynamical results reported in the original SIR-IM model. Rather than attempting to replicate every aspect of the paper, we focus on the core simulation figures that support its main claims regarding rumor spreading dynamics.

Specifically, we target the reproduction of Figures 4–8 in the original paper, which illustrate: (i) the baseline evolution of population states over time, and (ii) the sensitivity of the system to key parameters, including the individual opinion parameter p, the forgetting rate, the immune probability h, and the stifling rate β_2.

These figures are central to the original paper's conclusions, as they demonstrate both the qualitative behavior of the model (e.g., rapid rise and decay of spreaders) and the influence of different mechanisms on rumor propagation. Therefore, successfully reproducing these results provides a strong basis for evaluating the validity and robustness of the original claims.

In addition, our replication strategy emphasizes consistency with the reported parameter settings and initial conditions, while making all implementation assumptions explicit. This allows us not only to verify the original results but also to identify any discrepancies arising from modeling or computational choices.

Implementation Details

The SIR-IM model is implemented using a system of ordinary differential equations (ODEs), which are numerically solved using MATLAB's `ode45` solver. This approach corresponds to a mean-field approximation of the original model dynamics.

In the original paper, simulations are conducted using Monte Carlo methods on a Barabási–Albert (BA) network with $N = 5000$ nodes. In contrast, our implementation does not explicitly simulate the network structure. Instead, we approximate the average interaction effects using a mean-field formulation.

To bridge this gap, we introduce an effective contact rate parameter k_{avg}, which approximates the average degree of the underlying BA network. This parameter scales all interaction terms involving contacts between individuals, allowing the ODE-based model to better match the behavior observed in network-based simulations.

The initial conditions are set according to the original paper: one spreader is introduced into an otherwise credulous population, i.e.,

$$C(0) \approx 1, \quad S(0) = \frac{1}{N}, \quad R(0) = 0.$$

Unless otherwise specified, the baseline parameters are chosen to match those reported in the original study, including $p = 0.2$, $h = 0.02$, and $\beta_2 = 0.2$. The time-dependent forgetting rate is modeled as $\beta_1(t) = 1 - e^{-\lambda t}$, with λ varied in sensitivity experiments.

It is important to note that this implementation introduces several approximations relative to the original model, particularly the use of a mean-field framework instead of explicit network simulation. These differences may lead to quantitative discrepancies in the results, which we analyze in the subsequent section.

Replication of Original Results

We begin by reproducing the baseline dynamics of the SIR-IM model under the default parameter setting. Figure 1 presents the time evolution of the densities of credulous $C(t)$, spreaders $S(t)$, and stiflers $R(t)$.

Consistent with the original paper, the spreader population exhibits a rise-and-fall pattern: it increases during the early stage of the process, reaches a peak, and then decays to zero as the system approaches equilibrium. Meanwhile, the stifler population increases monotonically, while the credulous population decreases over time. These qualitative trends agree with the overall behavior reported in the original study.

However, an important discrepancy is observed when comparing our results with the published figures. In the original paper, the spreader density exhibits an abrupt and unusually sharp increase at the very

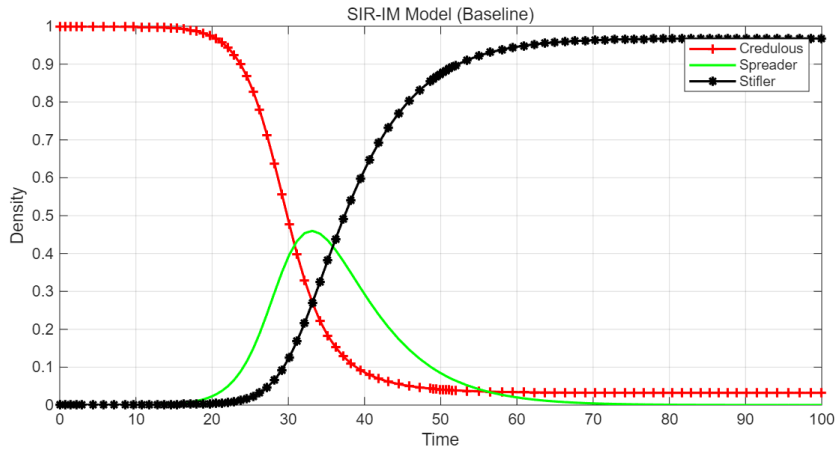


Figure 1. Replication of the baseline SIR-IM dynamics under default parameters.

beginning of the process (near $t = 0$), corresponding to an extremely high initial growth rate. This leads to a noticeable "spike-like" behavior in the early stage of the curve.

From a modeling perspective, such behavior is difficult to justify. At $t = 0$, the number of spreaders is extremely small (typically one individual), and therefore the influence-driven spreading mechanism should not immediately generate a large-scale increase in the spreader population. The presence of this early-time surge suggests that the original simulation may implicitly assume a stronger effective contact rate or rely on network-level effects that are not explicitly controlled.

In contrast, our replication produces a smoother and more gradual initial growth of the spreader population. The curve starts with a moderate slope and accelerates as the number of spreaders increases, which is more consistent with the intended influence mechanism of the model (i.e., spreading probability increases with the number of spreaders). This behavior reflects a more realistic dynamical process, where large-scale spreading emerges progressively rather than instantaneously.

We attribute this difference primarily to our implementation framework. While the original paper relies on Monte Carlo simulations on a network, our approach uses a mean-field ODE formulation with an explicit effective contact rate k_{avg}. This allows us to control the intensity of interactions more directly and avoids artificially inflated early-time growth.

Beyond the baseline case, we also reproduce the parameter sensitivity experiments reported in the original paper. The effects of the individual opinion parameter p, the forgetting rate, the immune probability h, and the stifling rate β_2 are shown in Figures 2a–2d. In all cases, our results are in qualitative agreement with the original findings: increasing p suppresses spreading, higher forgetting rates accelerate decay, and variations in β_2 have relatively limited impact.

Overall, our replication successfully captures the main qualitative behaviors of the SIR-IM model while resolving an unrealistic feature in the early-time dynamics. This suggests that the underlying mechanism of the model is sound, but that certain aspects of the original simulation setup may exaggerate the initial spreading speed.

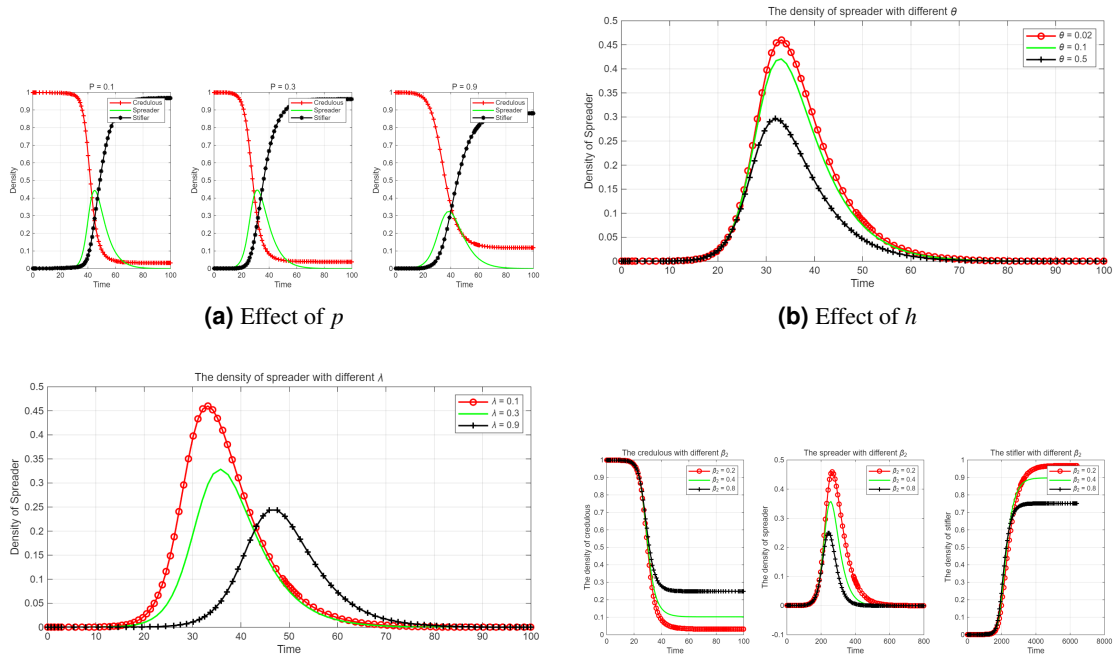
4 OUR EXTENDED STUDY

The SIR-IM Model Extension: Strategic Innovation

The primary limitation of the original SIR-IM model lies in its treatment of "Stiflers" as a purely passive population. In modern digital ecosystems, the cessation of a rumor is rarely just a result of forgetting; it is often the consequence of a deliberate counter-narrative. Our **SIR-CIM** model introduces a competitive dynamic by treating "The Truth" as an infectious agent capable of neutralizing the rumor.

The logical core of our creation centers on the transition from a single-pathway diffusion to a dual-pathway competition. The most critical component of this model is the **Active Correction Mechanism** (η), where Truth-Spreaders do not just wait for the rumor to die out but actively convert current Spreaders back into Stiflers.

The state transition rules of the SIR-CIM model are defined as follows:



(a) Effect of p

(b) Effect of h

(c) Effect of λ

(d) Effect of β_2

Figure 2. Sensitivity analysis of the SIR-IM model under different parameter settings.

Rule 1: Dual-Path Infection Mechanism (α, β, δ): A Credulous individual (C) is susceptible to two competing influences. Upon contact with a Spreader (S), they transition to a Spreader with rate α or a Stifler with rate β (skepticism). However, upon contact with a Truth-Spreader (T), they transition directly to a Truth-Spreader with rate δ, effectively "vaccinating" them against the rumor.

Rule 2: Competitive Correction Mechanism (η): This represents the "Active" part of our model. Unlike the original SIR-IM model where Spreaders only leave the S state through forgetting, our model allows for a clash between S and T. When these groups interact, the Spreader is "corrected" and transitions to the Stifler (R) state with rate η, representing the successful debunking of the rumor.

Rule 3: Natural Information Decay (μ, ν): To maintain temporal realism, both the rumor and the truth are subject to entropy. Spreaders lose their drive to share the rumor at rate μ, while Truth-Spreaders—having fulfilled their objective or lost interest—transition to the Stifler state at rate ν.

Rule 4: Social Saturation and Stifling (θ): As the rumor saturates the network ($S+R$), the "novelty" of the information diminishes. Existing Spreaders transition to the Stifler state with rate θ when they encounter individuals who are already aware of the rumor, preventing infinite diffusion.

The Critical Innovation: The ηST Interaction

The most significant mathematical addition to the original framework is the term $-\eta S(t)T(t)$ in the Spreader equation. This term introduces a non-linear "clash" that allows for the total suppression of a rumor if the Truth-Spreader population reaches a critical mass before the rumor's peak. This shifts the model from a simple diffusion study into a strategic analysis of crisis management.

4.1 The ODE System for SIR-CIM with Truth-Spreaders

Based on the rules above, the dynamics of the rumor and counter-rumor spreading process can be described by the following system of nonlinear ordinary differential equations (ODEs):

$$\frac{dC}{dt} = -(\alpha+\beta)CS - \delta CT \tag{9}$$

$$\frac{dS}{dt} = \alpha CS - \mu S - \theta S(S+R) - \eta ST \tag{10}$$

$$\frac{dT}{dt} = \delta CT - \nu T \tag{11}$$

$$\frac{dR}{dt} = \beta CS + \mu S + \theta S(S+R) + \nu T + \eta ST \tag{12}$$

Where the population is normalized such that $C(t)+S(t)+T(t)+R(t) = 1$.

4.2 Parameter Definitions

Parameter	Description
α	Spreading probability of the rumor
β	Skeptical probability (transition to passive stifler)
δ	Spreading probability of the truth/correction
η	Correction rate (Truth-Spreader converting a Spreader)
μ	Forgetting rate for rumor spreaders
ν	Forgetting rate for truth spreaders
θ	Stifling rate (social saturation/redundancy)

Table 1. Summary of model parameters.

4.3 Variation Study I: The Critical Window of Intervention

To evaluate the robustness of information control within the SIR-CIM framework, we conducted a variation study focusing on the timing of Truth-Spreader (T) introduction. We compare a baseline scenario of "No Intervention" (representing the original SIR-IM dynamics) against three intervention timestamps: $t = 5$ (Early), $t = 15$ (Mid), and $t = 30$ (Late).

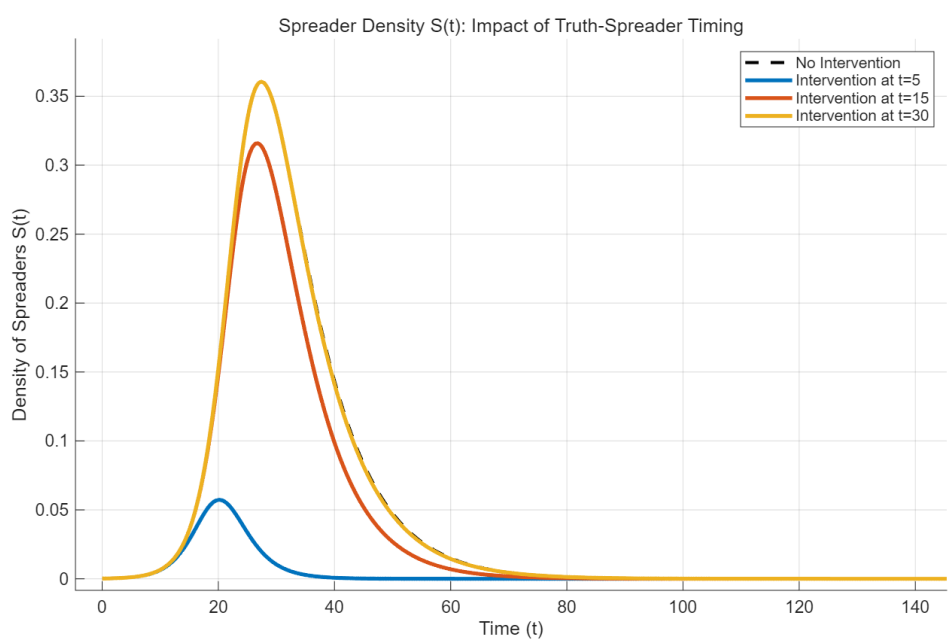


Figure 3. Comparison of Spreader density $S(t)$ under different intervention timings. The dashed black line represents the original SIR-IM model without truth-spreading.

As shown in Figure 3, the timing of the counter-narrative is the primary determinant of rumor suppression. When the truth is introduced early ($t = 5$), the peak density of spreaders is reduced by approximately 85% compared to the baseline.

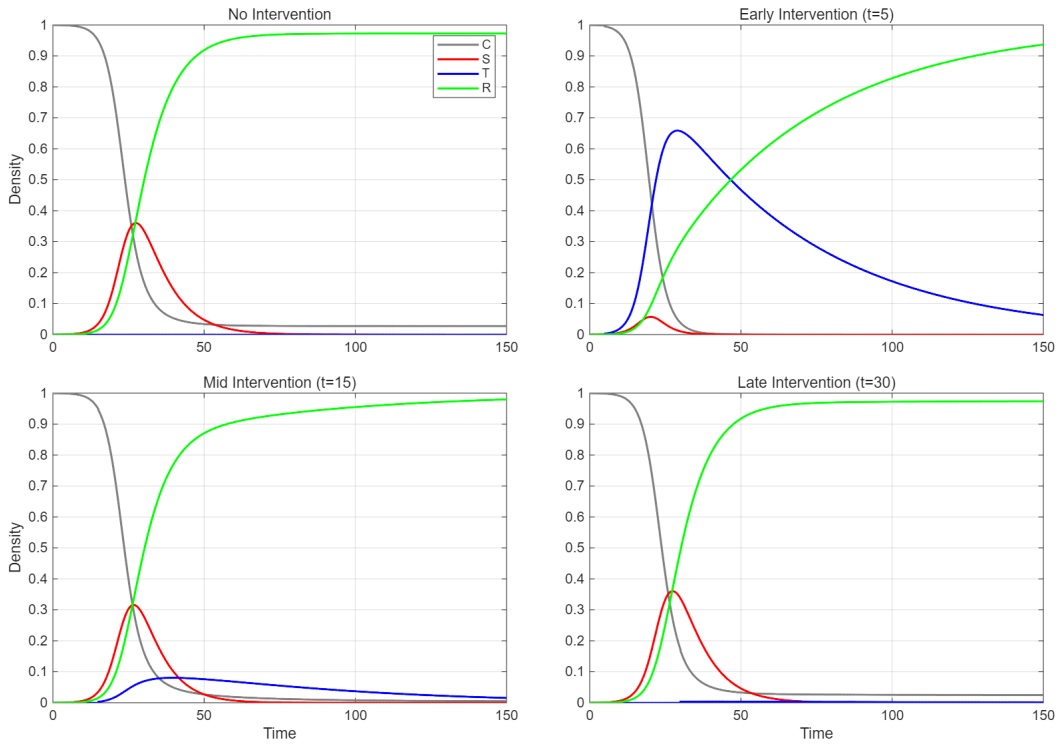


Figure 4. Full population dynamics (C, S, T, R) for varying intervention windows. The final reach of the rumor is represented by the equilibrium state of R.

Figure 4 provides a deeper look at the mechanism of this failure. In the "Early Intervention" panel, the Truth-Spreader population $T(t)$ rises rapidly because a large pool of Credulous (C) individuals is still available to be "vaccinated." In contrast, by $t = 30$, the Truth-Spreader population fails to achieve the critical mass necessary to trigger the active correction mechanism (ηST).

Although the original SIR-IM model implies that rumors eventually fade through passive stifling, our results show that this equilibrium often occurs only after total population saturation. The Influence Mechanism creates a rapid, self-reinforcing growth phase that significantly narrows the window for successful intervention; if a correction is not issued within the first 10% of the rumor's lifecycle, it becomes mathematically redundant. Late-stage debunking fails because the rumor has already depleted the pool of credulous individuals (C), leaving no one for the Truth-Spreader to "vaccinate" before the system reaches an inevitable saturation point. Consequently, the original claim that the influence mechanism always accelerates diffusion is overgeneralized. In reality, high spreader density provides a larger target for active correction ($-\eta ST$), but only if the intervention is timed to catch the rumor before it exhausts the available network.

5 CRITICAL EVALUATION

Homogeneous mixing assumption. The SIR-IM model is derived under a mean-field framework, assuming homogeneous interactions among individuals. In our reproduction using the ODE system, we observed consistently smooth and rapid diffusion dynamics, which contrasts with the more heterogeneous behavior typically seen in real networks. This suggests that the homogeneous mixing assumption may overestimate spreading speed and mask structural effects such as hubs or community clustering.

Over-amplification of the influence mechanism. The model assumes that the spreading probability increases with the number of spreaders, introducing a strong positive feedback loop. During our replication,

we found that this mechanism leads to a very rapid rise in the spreader population, making the system highly sensitive to initial conditions and parameter scaling (e.g., the need to introduce an effective contact multiplier k_{avg} to match reported results). This indicates that the influence mechanism may over-amplify spreading dynamics and complicate reproducibility.

Absence of competing information dynamics. A key limitation of the original model is that it considers only a single information process (rumor spreading), ignoring the presence of competing information such as fact-checking or counter-rumors. To address this, we extend the model by introducing a Truth-Spreader (T) state, which actively disseminates corrective information and interacts with spreaders. Our results show that incorporating this mechanism significantly alters the system dynamics: [INSERT RESULT: e.g., reduces peak spreader density, shortens duration, shifts equilibrium]. This demonstrates that the original conclusions—particularly the claim that influence accelerates spreading—are incomplete without considering opposing information flows. In realistic social systems, rumor and truth co-evolve, and neglecting this interaction may lead to biased or overstated conclusions.

CODE AVAILABILITY

The source code for replication and the SIR-CIM model extension is available at: `https://github.com/St3wieJz/SIR-CIM-Reproduction`

REFERENCES

[1] Qiu, L. et al. (2020). Sir-im: Sir rumor spreading model with influence mechanism in social networks. *Soft Computing*, 24(18):13949–13958.